Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local CSV File Dependency	The initial loan_prediction.csv data is static and relies on local storage rather than a live database.	High
2	Single-threaded Flask Server	The default Flask development server is not built to handle thousands of concurrent application submissions.	High
3	Static Feature Inputs	User must manually enter credit and income details, which could be error-prone or falsified.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handled by local Flask app server.	Deploy the Flask application to a cloud provider (AWS/Azure) using a production WSGI server (Gunicorn) and load balancer.
2	Data Storage	Relies on .csv files and local disk storage.	Migrate to a relational cloud database (like PostgreSQL or MySQL) to manage applicant profiles and loan requests dynamically.
3	Performance	.pkl models are loaded via standard disk read.	Cache the serialized models in memory on app startup to prevent disk I/O bottlenecks during peak traffic.
4	Security	Basic input validation on frontend.	Implement JWT-based authentication for bank officers and SSL/TLS encryption for applicant financial data transmission.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Database Migration & API Refactoring	Q3 2026	Replaces CSV files with a robust PostgreSQL database, enabling dynamic reporting and application tracking.
Phase 3	Third-Party Credit Score Integration	Q4 2026	Integrates external APIs to automatically fetch accurate credit history instead of relying on manual applicant input.
Phase 4	Model Retraining Pipeline	Q1 2027	Implement an automated pipeline that periodically retrains the ML model on new applicant data to maintain prediction accuracy over time.